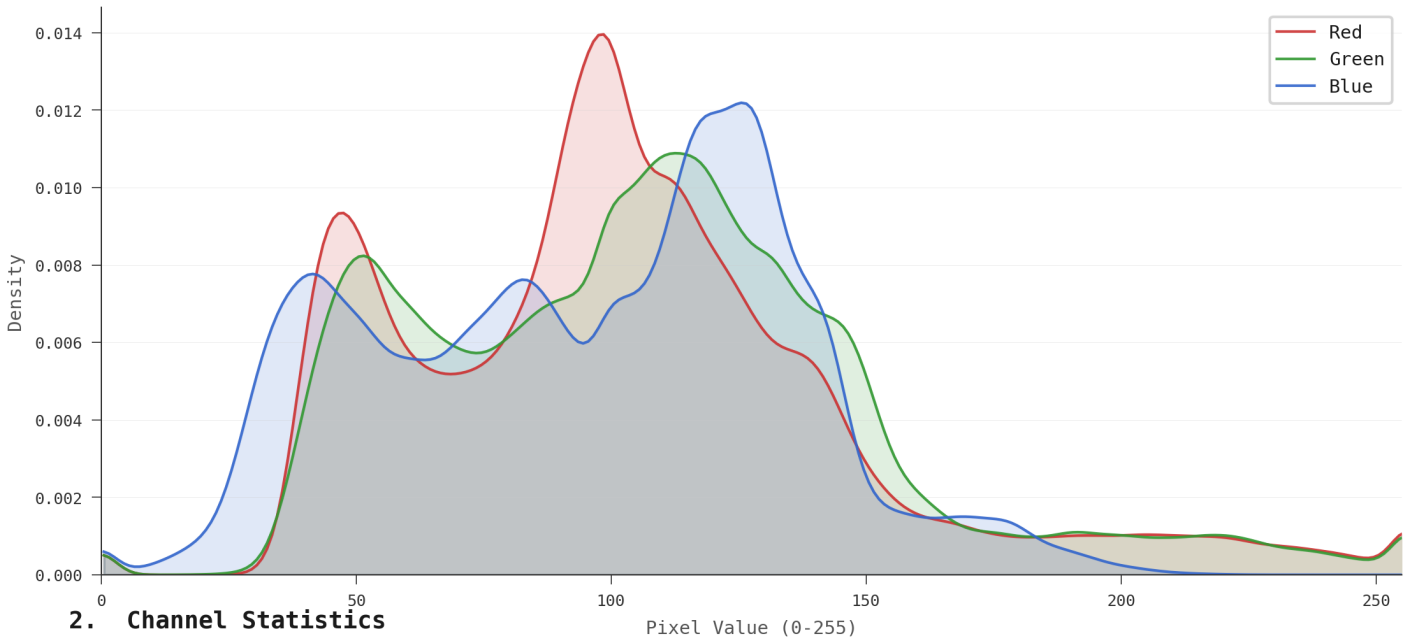
Reference Data Sheet: KODIM16

Subject: Island Panorama Aerial

768x512 | 393,216 px
24-bit RGB | sRGB | PNG (lossless)

1. RGB Channel Distribution



2. Channel Statistics

Channel	Mean	Std Dev	Variance	Kurtosis	Skewness	Min	Max
Red	102.8127	43.8151	1919.7670	1.1627	0.9143	0	255
Green	106.6684	44.0190	1937.6721	0.6672	0.6923	0	255
Blue	95.1600	39.7373	1579.0502	-0.7313	-0.0325	0	249

3. Inter-Channel Correlation

	R	G	B
R	1.0000	0.9842	0.8923
G	0.9842	1.0000	0.9511
B	0.8923	0.9511	1.0000

Pairwise Pearson Correlation Coefficients

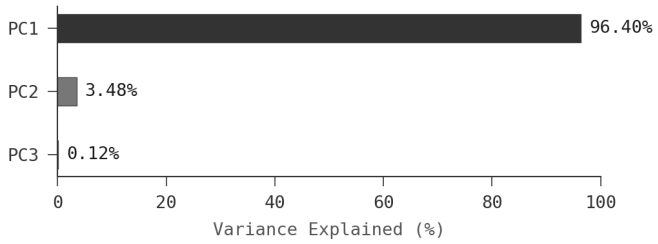
r(R-G)	=	**0.984232**
r(R-B)	=	**0.892330**
r(G-B)	=	**0.951070**
Avg |r|	=	**0.942544**

4. Covariance Matrix S

$$
\begin{array}{c} \\ R \\ G \\ B \end{array}
\begin{array}{c}
\begin{array}{ccc} R & G & B \end{array} \\
\begin{bmatrix}
\mathbf{1919.7718} & 1898.2915 & 1553.6340 \\
1898.2915 & \mathbf{1937.6771} & 1663.6101 \\
1553.6340 & 1663.6101 & \mathbf{1579.0542}
\end{bmatrix}
\end{array}
$$

Reference Data Sheet: KODIM16

5. Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.5939	**-0.6071**	-0.5278	5240.9414
PC2	-0.6045	-0.0962	0.7908	189.2029
PC3	0.5309	-0.7887	0.3099	6.3588

6. Derived Metrics

Condition Number

824.20

(l1/l3) PC1/PC2: 27.70 | PC2/PC3: 29.75

Blue Channel Independence

7.5%

Fraction of blue variance not captured by PC1

PC1 Dominant Channel

Green

Loading: -0.6071 | Green dominant

Classification: Strongly One-Dimensional (PC1 = 96.40%)

7. Spatial Autocorrelation (lag-1)

Channel	Horizontal	Vertical	Average
Red	0.983700	0.943618	0.963659
Green	0.983997	0.945143	0.964570
Blue	0.981362	0.935445	0.958404

8. Average Local Variance (3x3)

Channel	Value
Red	103.5563
Green	101.1993
Blue	94.5820

9. Redundancy Profile

Classification: Strongly One-Dimensional | Condition Number: 824.20 | PC1: 96.40%

Correlation: R–G pair dominates at r = 0.984232 | G–B: 0.951070 | R–B: 0.892330

Blue Channel: 7.5% independent | Moderate decoupling from PC1

Spatial: Avg autocorrelation: 0.962211 | Avg local variance: 99.7792

Eigenvector: Green dominant | G (-0.6071) leads | R (-0.5939), B (-0.5278)

References

[1] Eastman Kodak Company. Kodak Publication No. PCD-042, 1992.

[2] Baetzel, J. (2026). "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."

[3] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.

[4] Giorgianni, E.J. and Madden, T.E. Digital Color Management. Addison-Wesley, 1998.

Blue Channel Independence = (1 - (blue_loading_PC1^2 x l1 / Var(B))) x 100

Data Source: Statistics derived from Baetzel (2026), "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."